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# Greenhouse gas (GHG) emissions reduction in the electricity sector: Implications of increasing renewable energy penetration in Ghana's electricity generation mix

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ABSTRACT

The development and deployment of renewable energy technologies into the national electricity generation mix have high potential in the reduction of greenhouse gas (GHG) emissions in the sector, which contributes to about 40% of GHG emissions globally. Ghana through the Nationally Determined Contributions has pledged to reduce GHG emissions by 15% unconditionally, especially in the electricity sector which is one of the major contributors of GHG in the country. The study explores how the impact of increasing renewable energy penetration in the electricity generation mix will have on the greenhouse gas (GHG) emissions reduction from the sector. Four electricity supply scenarios were developed in the Low Emissions Analysis Platform (LEAP) using existing renewable energy targets identified in Ghana's Renewable Energy Master Plan. These, including, the Business as Usual (BAU), 10%, 20% and 30% renewable energy penetration scenarios were modelled to assess GHG emissions reduction up till 2030 by replacing the thermal generation sources in the electricity generation mix with equivalent amounts of renewable energy sources. Electricity demand projections for the various consumption sectors including residential, non-residential, special load as well as street lighting were made from 2016 to 2030 based on historical data. The model projected the demand for electricity to rise from 10,129.0 GWh in 2016 to 24,151.4 GWh in 2030. The cumulative demand of 237,570.5 GWh was met in all four scenarios with a total installed generation capacity of 3,794.6 MW. GHG emissions for the BAU scenario rose cumulatively to 52,545.2 thousand $MtCO_2eq$ in 2030 but reduced to 46,178.3 thousand $MtCO_2eq$, 41,603.2 thousand $MtCO_2eq$ and 38,332.6 thousand $MtCO_2eq$ in the 10%, 20%, and 30% renewable energy penetration scenarios respectively. The result indicates that renewable energy deployment could lead to the attainment of GHG emission reduction substantially leading to minimising effects on climate change.

## Introduction

Although energy is crucial in any economy's growth and development, its production and use contribute significantly to growing global environmental challenges including climate change. Fossil fuels have been identified as the major contributor to greenhouse gas

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(GHG) emissions in the energy sector [1,2]. In 2019, fossil-based fuels including coal, natural gas and crude oil together accounted for more than 60% of the total electricity generated [3]. The increasing demand for fossil fuel in the electricity generation sector has further worsened the situation of GHG (mainly carbon dioxide) emissions amidst carbon capture and sequestration technologies (to minimize carbon dioxide emissions), which are at their early stages of research and development [4]. Chanda and Bose [5] opined that the influence of these global environmental challenges would be severe in the tropical regions where most developing countries are located.

Due to the rise in the demand for electricity produced mainly from fossil fuels, the carbon dioxide concentration in the atmosphere has significantly increased, leading to an observed increase in average atmospheric temperatures [6]. The national oceanic and atmospheric administration (NOAA) reported the year 2019 as the second warmest year in history with $a + 0.95$ °C increase in global land and ocean surface temperatures, next to 2016 with $a + 0.99$ °C increase in global land and ocean surface temperatures. According to British Petroleum [7], carbon dioxide emissions rose from 29,714.2 million tonnes in 2009 to 33,444.0 million tonnes in 2017.

The Intergovernmental Panel on Climate Change [8] also reported that the African continent is especially vulnerable to the impacts of climate change due to factors such as widespread poverty, prolonged droughts, unequal land distribution and overreliance on rain-fed agriculture. The report further concludes that, although adaptation strategies, including traditional coping mechanisms, are theoretically possible, the infrastructural and economic response capacity to effect timely responses may well be beyond some countries' economic means. Yadoo & Cruickshank [9] noted that if developing countries follow the same paths of growth as the industrialised nations of today, there would be a sharp increase in global GHG emissions. Since electricity generation has been reported to account for about 40% of GHG emissions globally [4], Owusu & Asumadu-Sarkodie [10] highlights the need to build international collaborations that supports developing countries towards the accessibility of renewable energy and also promote new potential towards climate change mitigation.

The consequences of the global environmental crises are already manifesting in Ghanaian and many other developing countries' societies and economies. The WorldBank [11] reports that increased climate variability already affects Ghana's economy and households, especially in vulnerable sectors such as agriculture, coastal zones / marine ecosystems, water supply and energy production. If nothing is done to lessen the effect of climate change, it is estimated that the annual real GDP for Ghana would be 1.9% to 7.2% lower than it should have been by 2050 [11]. Therefore, Ghana has made some commitment through Nationally Determined contributions by ensuring that by 2030, Ghana will reduce emissions by 64 MtCO2e in absolute terms by implementing 34 mitigation initiatives. Nine unconditional initiatives out of the 34 are anticipated to reduce CO2 emissions by 24.6 MtCO2e. If sufficient financial assistance from the international and private sectors is made available to cover the full cost of implementation, an additional 25 conditional measures can be put into place to further attain 39.4 MtCO2e.

Planning adaptation or preventive measures to address the impact of global climate change resulting from increasing GHG emissions is therefore imminent to reducing the consequences of climate change and improving the future environment [12]. Timmons et al. [13] recommended a shift from the over-reliance on fossil fuels to increasing renewable energy usage in order to tackle environmental issues, fuel supply and price restrictions, and technological change.

Many countries, especially the developed ones, are investing in renewable energy, such as solar, wind, hydropower or biomass as a result of the harmful and life-threatening consequences of climate change. The proportion of renewable energy (RE) in the global electricity generation mix increased to 27.3% in 2019 [3]. Out of this figure, hydropower contributed more than half whiles wind, solar, biomass and other RE systems accounted for a little less than 12% of the global electricity generation mix [3]. This transition to "cleaner energy" is expected to result in the reduction of GHG emissions; thereby helping to reduce impact of climate change. Squalli [14] stipulated that renewable energy is widely seen as a tool in reducing environmental pollution, enhancing public health, stimulating economic development through the creation of employment and providing a more efficient and sustainable energy network. It is for this reason that Ghana, in its commitment to reducing GHG emissions, set emission reduction goals to reduce its GHG emissions unconditionally by 15% by 2030 (United Nations Framework Convention on Climate Change [[15]CC], 2015). To achieve this aim, the country is seeking to expand its renewable energy sector to complement the generation mix of 57.10% — Thermal, 42.70% — Hydro and 0.20% — Renewables as at the end of December 2016 [16]. The low production of renewable energies in Ghana over the years can be attributed to obstacles ranging from technological, regulatory, financing, and project implementation. Mahama et al. [17] suggested that the production of renewable energy is an important element in the growth of Ghana's energy sector, especially to substitute fossil fuels that lead to climate change through GHG emissions.

The literature in the African context and for that matter Ghana, have mainly focused $CO_2$ from energy consumption and economic growth and their relationship (see [18–27]). These studies using different methodologies suggest that energy consumption and economic growth have a significant impact on carbon dioxide emissions in Africa. Al-mulali & Sab [19] found that energy consumption has played an important role in increasing economic growth and financial development in Sub Saharan African countries, but with the consequence of high pollution. Khobai [23] found that there is a long-run relationship between energy consumption, carbon dioxide emission, and economic growth in South Africa. Overall, the papers suggest that African countries need to focus on improving energy consumption efficiency and alternative energy resources, which produce or emit less GHG primarily $CO_2$ to achieve economic growth without adversely affecting the environment. Whereas Nyansapoh, Debrah, Anku & Yamoah [28] analysed the electricity generation system and the impact of fuel options (fossil fuel electricity generation plants) on the environment in Ghana, their study did not demonstrate how renewable energy resources can help reduce GHG emissions and this paper fills that gap using LEAP, a widely recognized and validated modelling tool, enhances the credibility and robustness of findings. It allows for a systematic and data-driven evaluation of the impact of renewable energy integration.

On both local and global fronts, efforts are being made to minimize GHG emissions from the electricity generation sector by increasing the penetration of renewable and other alternative energy sources into the generation mix. The objective of this paper is to analyse how the impact increasing penetration of renewable energy sources in Ghana's electricity generation mix will have on overall electricity supply as well as GHG emissions reduction; towards achieving the intended nationally determined contribution (INDC) of 15% by 2030, in consonance with the argument by Twidell & Weir [29] that the replacement of renewable energy with fossil fuels reduce the emission of GHG significantly in the electricity generation sector. The research specifically targets Ghana, an African country. This localized focus allows us to provide tailored recommendations and insights for the Ghanaian context. The study also incorporates Ghana's Renewable Energy Master Plan and Energy Policy targets, ensuring that our scenarios align with the country's existing renewable energy targets and policies. This integration facilitates the practical implementation of the research findings within the country's policy framework.

The implementation of the research finding will contribute the achievement of SDG

## Electricity demand and supply mix in Ghana

Electricity consumption is classified under residential, non-residential, special load (high, medium and low voltage categories) and street lighting consumers. Special load consumers refer to industrial consumers who are mostly classified under low, medium or high voltage were the biggest consumers of electricity over the last decade as shown in Fig. 1. Residential consumers were the next biggest consumers of electricity, with non-residential which is mostly for commercial facilities coming in third.

Electricity demand in Ghana has seen a steady rise over the last decade, increasing from 1423 MW in 2009 to 2525 MW in 2018 [30]. Over the same period, installed generation capacity increased by over 250% from 1970 MW in 2009 to 4889 MW in 2018 as shown in Figure SM1(supplementary material) [30,31]. This suggests that demand growth was not taken into consideration in the planning and installation of generation facilities from 2013 till 2018, since the installed generation capacity as at 2018 was double the demand in the same year. Inspite of this increase in installed capacity, the country has seen its fair share of electricity crises resulting from irregular supply of fuel.

The electricity generation mix in Ghana has seen significant transformation over the last decade moving from heavy dependence on hydro sources to various thermal sources. Renewable energy has not played any significant role in Ghana's generation mix. Fig. 2 shows that the proportion of hydro power in the generation mix of Ghana has reduced from about 75% in 2009 to about 35% in 2018. Thermal generation on the other hand increased from 25% in 2009 to 65% in 2018. Electricity production from thermal sources is mostly from natural gas, heavy fuel oil, light crude oil and diesel.

The contribution of renewable energy to the generation mix is not significant enough to make it visible in Fig. 3. As at the end of 2018, the total installed capacity from renewable energy sources was 42.6 MW comprising of 42.5 MW large-scale solar photovoltaic plants and 0.1 MW biogas power generation plant. This journey began with a 2.5 MWp solar photovoltaic facility operated by the volta river authority (VRA), which happened to be the first large-scale renewable energy facility integrated into the generation mix of Ghana. The installed renewable energy generation capacity increased from 2.5 MW in 2013 to 22.6 MW at the end of 2016 and subsequently to 42.6 MW at the end of 2018 [30]. The government of Ghana targeted 10% of the electricity generation mix to come from renewable energy sources by 2020 but has since updated this target to 2030 in the Renewable Energy Master Plan developed by the Energy Commission of Ghana [32].

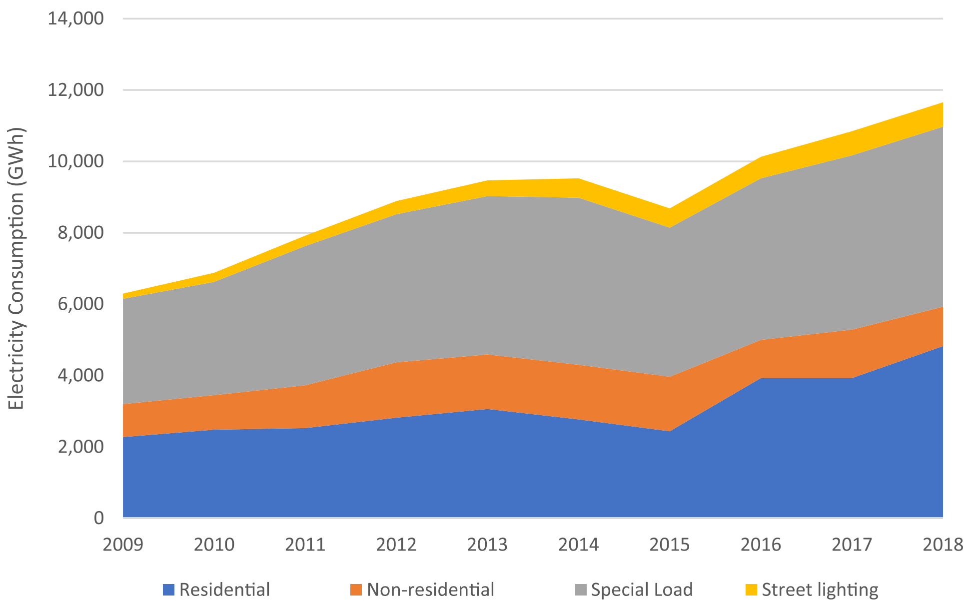



**Fig. 1.** Electricity consumption by categories from 2009 to 2018.

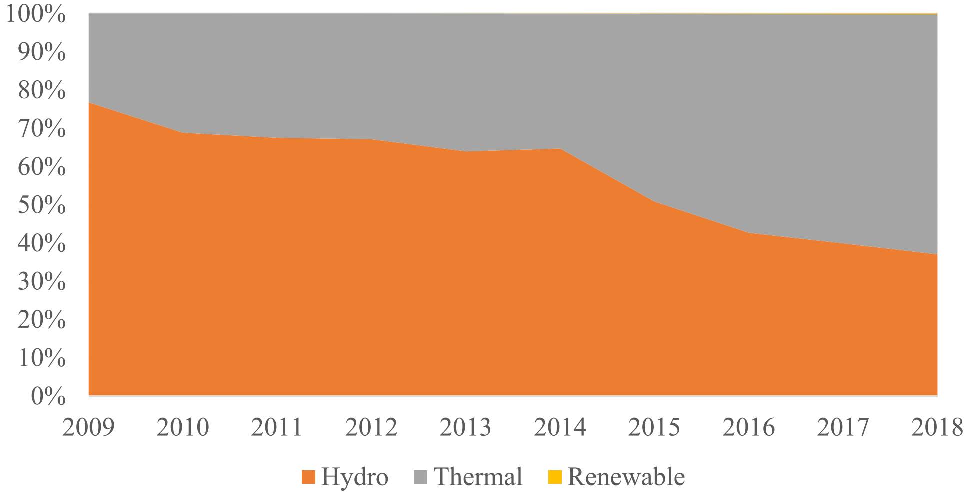



**Fig. 2.** Electricity Generation Mix from 2009 to 2018.

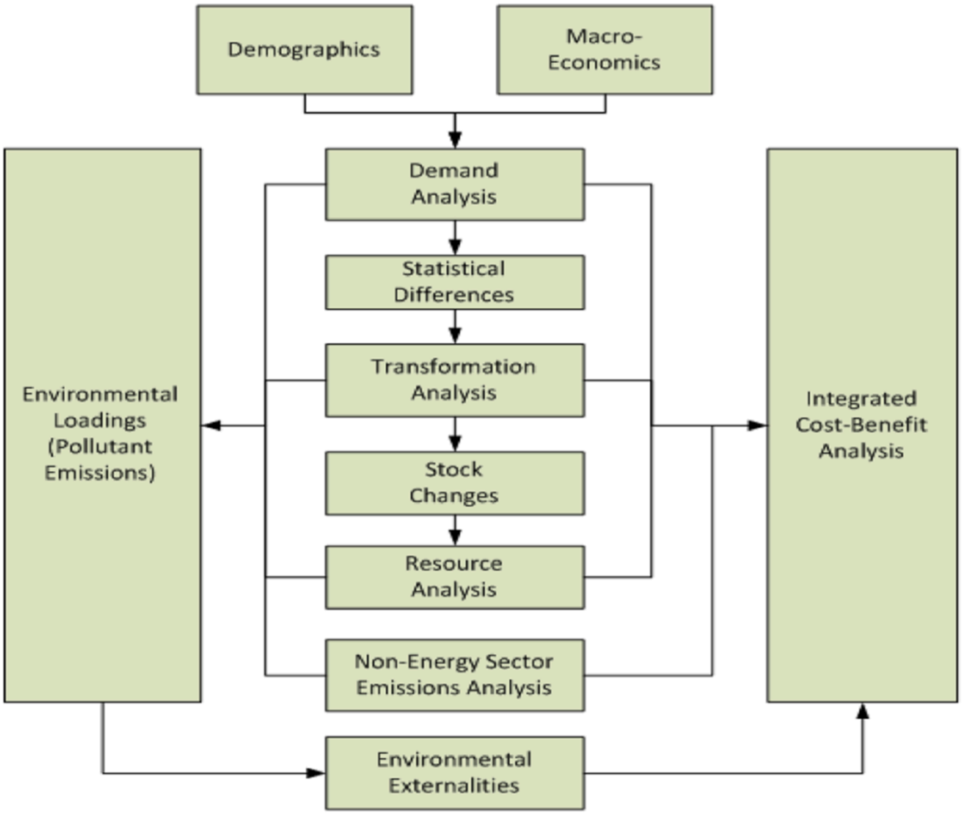



**Fig. 3.** The Structure of LEAP Calculations [33].

## Methodology

### *Study scope and analysis period*

The study comprises electricity demand projections, matching electricity supply options and GHG emissions analysis associated with electricity supply. Models were developed to project electricity demand and supply as well as estimate the GHG emissions associated with electricity generation. The base and end years for the model were 2016 and 2030 respectively. The final energy demand analysis was carried out over the period (2016 – 2030) using historical data going as far back as 2000. Demand projections were conducted for all demand sectors including residential, non-residential, special load and street lighting. On the supply side, electricity generation facilities utilising hydro, thermal and renewable energy sources available as of 2016 were modelled using data from the Energy Commission of Ghana. Supply projections were conducted in four (4) scenarios with different levels of renewable energy penetration and their associated GHG emission reductions estimated. These were carried out with the help of the low emissions analysis platform (LEAP).

*Model structure and scenarios*

The low emissions analysis platform (LEAP) is a software tool, developed by Sweden's Stockholm Climate Institute and the University of Boston in the United States of America, for energy policy, climate change mitigation and air pollution decline forecasting. It is an integrated modelling tool that can be used to track energy consumption, production and resource extraction in all sectors of an economy. It can be used to account for both energy sector and non-energy sector greenhouse gas (GHG) emission sources and sinks, in addition to tracking GHG. It has the capability to forecast medium to long-term energy demand and supply under different conditions for whole societies and measures the amount of GHG emitted in the energy distribution and usage cycle. The tool has the capacity to undertake resource analysis in order to ascertain the sufficiency of energy resources in a society.

LEAP can be used to generate models of diverse energy systems, where each needs its own unique data sets and also supports a variety of different modelling methodologies on both the demand and supply sides. On the demand side these range from ascending, end-use accounting techniques to descending macroeconomic modelling whiles LEAP runs a range of accounting, simulation, and optimization methodologies, On the supply side, for modelling electricity generation and capacity expansion planning. It is also easy to allow LEAP to incorporate data and results from other more specialized models. The modelling capabilities of LEAP operate at two basic conceptual levels; built-in calculations handle all energy, emissions and cost-benefit accounting calculations in one methodology while in the other methodology, users enter spreadsheet-like expressions that can be used to specify time-varying data or to generate a wide variety of sophisticated multi-variable models, thus enabling econometric and simulation approaches to be embedded within LEAP's overall accounting and optimization frameworks. LEAP is designed around the concept of long-range scenario analysis. This paper utilized the built-in calculations method and takes demographic, macro-economic as well as historical electricity demand and supply data to project electricity demand and supply over the medium to long term as shown in Fig. 3.

*Electricity demand analysis*

Final electricity demand analysis and modelling in Ghana were projected using a disaggregated, end-use approach. The end-use demand data required for modelling final electricity consumption up to 2030, was disaggregated in the residential, non-residential, special load tariff and street lighting categories. In estimating the electricity demand, the total population with access to electricity (total activity) and the energy intensity of the population is required. The activity level analysis was the type of methodology that was adopted under the demand analysis. The final energy demand in which energy consumption is calculated as the product of an activity level and energy intensity is obtained from Eq. (1) [34].

$$D_{b,s,t} = TA_{b,s,t} \times EI_{b,s,t} \tag{1}$$

Where:

*D*: electricity demand
*TA*: total activity
*EI*: energy intensity
*b*: branch
*s*: scenario
*t*: year

Electricity demand projections for the Residential, Non-Residential, Special Load Tariff (SLT) and Street Lighting sectors were carried out from 2016 to 2030, using historical data (2000 – 2016). The average annual growth rates of electricity in the residential, non-residential and special load sectors estimated from historical data were 7.2%, 5.5% and 2.4% respectively. In the case of the streetlight, the overall annual growth rate was 21%, but an actual growth rate of 16.6%, representing a historic growth rate excluding 2005 - 2006 as well as 2009 – 2010, was used. The growth rates of 69.4% between 2005 and 2006 and 43.5% between 2009 and 2010, considered outliers, can be attributed to intensified street lighting exercises embarked on by the government of Ghana.

*Electricity supply projections*

Ghana's electricity generation sector has seen significant growth over the last decade. This growth was accompanied by shift from dependence on hydro to reliance on thermal sources namely natural gas, light crude oil, heavy fuel oil and diesel on some extreme cases. As at the end of 2016, the total installed generation capacity was 3785 MW with only 0.2% renewable energy penetration. Hydro and thermal sources for electricity generation accounted for 42.7% and 57.1% respectively. In this study, the electricity generation sector with all the installed generation plants available in 2016, was modelled without any policy interventions in the business-as-usual (BAU) scenario and projected to 2030. Three other scenarios were developed to cater for the different levels of renewable energy penetration in the generation mix. These include 10%, 20% and 30% renewable energy penetration as detailed in Table 1. These translate into 380 MW, 760 MW and 1138 MW for 10%, 20% and 30% renewable energy penetration respectively. The renewable energy sources include solar, wind, biomass, hydro (small, mini and micro) as well as waste-to-energy (W2E). The renewable energy penetration scenarios follow the National Energy Policy-2010 and the Renewable Energy Master Plan [32]. The projections took into account existing generation system particulars including installed capacity, fuel type, process efficiency and despatch order in the modelling process. Historical data on transmission and distribution losses were factored into the modelling process.

### *GHG emissions analysis*

Ghana, in its commitment to the united nations framework convention on climate change (UNFCCC), seeks to unconditionally reduce its GHG emissions by 15% by the year 2013 as part of its intended nationally determined contribution (INDC). In order to achieve this target, the country needs to diversify its electricity generation mix by including more renewable energy sources to replace some of its thermal sources. This is in line with the target in the National Energy Policy to increase the penetration of renewable energy in the national generation mix to 10% by 2020. The three scenarios developed in this study namely 10%, 20%, and 30% renewable energy penetration scenarios were modelled to assess GHG emissions and compared to the business-as-usual (BAU) in order to estimate the emission reductions to be achieved. The estimation of emissions were based on Eq. (2) [34].

$$Emission_{t,y,v,p} = EnergyConsumption_{t,y,v} \times EmissionFactor_{t,v,p} \times EmDegradation_{t,y-v,p} \tag{2}$$

Where:

*EmissionFactor*: the emissions rate for pollutant (p) from new power plant added in the year, v

*EmDegradation*: a factor representing the change in the emission factor for pollutant (p) as the power plant ages (It equals 1 when $y = v$. it is entered as a lifecycle profile).

t: technology type
y: calendar year
v: year technology was added
p: pollutant (GHG)

The 10% renewable energy penetration scenario follows the 2010 National Energy Policy target of 10% renewable energy penetration in the generation mix to replace an equivalent capacity of fossil-based fuel-powered plants by 2020. The 20% and 30% renewable energy penetration scenarios were modelled to determine the GHG emission reductions needed for Ghana to achieve its intended nationally determined contribution (INDC) to the United Nations Framework Convention on Climate Change (UNFCCC) to which Ghana is a party, by 2030. In these cases, fossil-based fuel-powered plants with a capacity equal to 20% and 30% of the installed base year generation capacity were removed by 2030 and replaced with renewable energy sources.

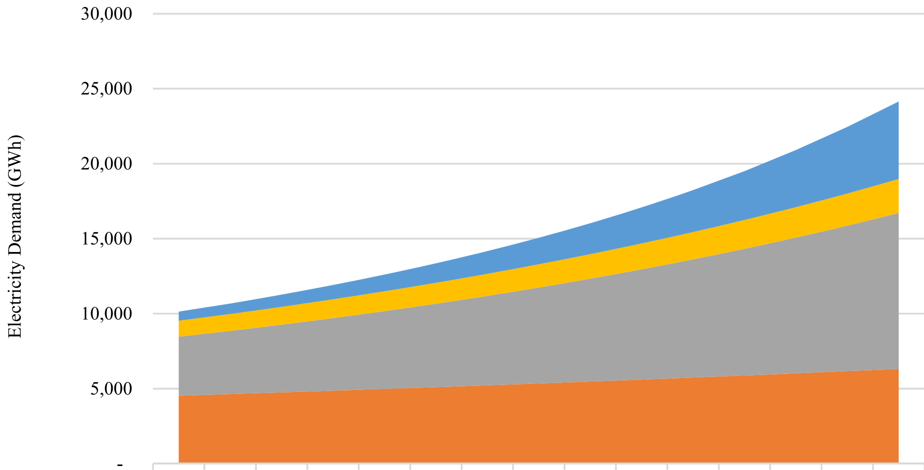
## Results and discussions

### *Electricity demand analysis*

Total electricity consumption is expected to increase from 10,129.0 GWh in 2016 to 24,151.4 GWh in 2030 as shown in Fig. 4. The electricity consumption in 2030 is about 2.4 times more than that of the base year. It is estimated that by 2019, demand for residential electricity will surpass that of the special load tariff sector. This growth can be attributed to the massive rural electrification programs embarked on in recent years by successive governments as well as the government's commitment to the objective of universal electricity access by 2020 in Ghana as outlined in the various government policies including the National Energy Policy, Sustainable Energy for All (SE4ALL) and more recently the renewable energy master plan (REMP). The demand for street lighting is expected to increase above that of non-residential electricity consumption by 2030, all things being equal. This is due to the increasing demand for the expansion of street lighting in communities due to increasing population and urbanisation. Non-residential consumption is

**Table 1**
Details of renewable energy sources in GHG emission reduction scenarios [32].

| Renewable Energy Sources | Business-As-Usual (BAU) (MW) | New Renewable Energy Plants to Replace Thermal Plants (MW) | | |
|---|---|---|---|---|
| | | 10% Renewable penetration | 20% Renewable penetration | 30% Renewable penetration |
| Solar | 42.5 | 180 | 360 | 541 |
| Wind | - | 100 | 200 | 325 |
| Biomass | - | - | 50 | 72 |
| Hydro (small, mini and micro) | - | 50 | 100 | 150 |
| Waste-to-Energy (W2E) | 0.1 | 50 | 50 | 50 |
| Total | 42.6 | 380 | 760 | 1138 |




| | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 | 2026 | 2027 | 2028 | 2029 | 2030 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Street Lighting | 603 | 703 | 820 | 956 | 1,115 | 1,300 | 1,515 | 1,767 | 2,060 | 2,402 | 2,801 | 3,266 | 3,808 | 4,440 | 5,177 |
| Non Residential | 1,066 | 1,125 | 1,186 | 1,252 | 1,321 | 1,393 | 1,470 | 1,551 | 1,636 | 1,726 | 1,821 | 1,921 | 2,027 | 2,138 | 2,256 |
| Residential | 3,932 | 4,215 | 4,519 | 4,844 | 5,193 | 5,567 | 5,967 | 6,397 | 6,858 | 7,351 | 7,881 | 8,448 | 9,056 | 9,708 | 10,40 |
| Special Load Tariff | 4,528 | 4,637 | 4,748 | 4,862 | 4,979 | 5,098 | 5,220 | 5,346 | 5,474 | 5,605 | 5,740 | 5,878 | 6,019 | 6,163 | 6,311 |

**Fig. 4.** Electricity Demand Projections for various consumption sectors in Ghana.

estimated to be the lowest of all the electricity consumption categories by 2030.

Adom et al. [35] identified the coefficient for industrial efficiency as negative and inelastic, thus a 1% increase in industrial efficiency saves 0.386% of aggregate domestic electricity consumption per annum. On the other hand, the income elasticity of electricity demand was identified as positive and elastic; meaning a 1% rise in real GDP per capita results in a 1.6% increase in aggregate domestic demand for electricity, thus the overall domestic demand for electricity is projected to grow with economic growth in Ghana [35]. This explains why residential power demand in this analysis has surpassed that of the special load in Fig. 4.

*Electricity supply projections*

Electricity generation in the BAU scenario did not include the addition of any new power plants, leaving the total installed generating capacity of 3794.6 MW. This installed capacity was sufficient to generate all of the electricity needed up to 2030 as shown in Figure SM2 (supplementary material), with specific details of the electricity production from the various generation facilities outlined in Tables SM1, SM2, SM3 and SM4 (supplementary material). The results show that the projected cumulative demand of 237,570.5 GWh of electricity from 2016 to 2030 was met in all four scenarios including the BAU. For each of these scenarios, total electricity generation for the period was 247,469.5 GWh, 247,426.35 GWh, 246,869.78 GWh for BAU, 10% renewable energy penetration, 20% renewable energy penetration and 30% renewable energy penetration scenarios respectively. The total generation in all four scenarios

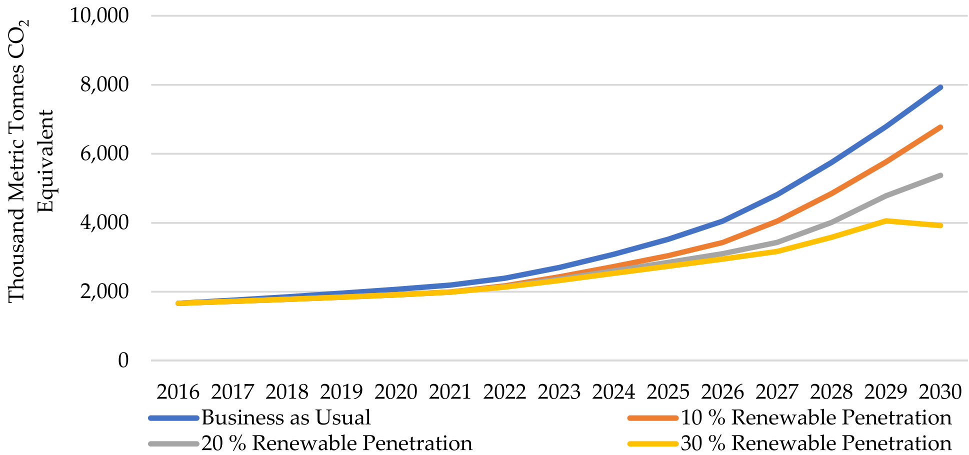



**Fig. 5.** GHG emissions at the point of emissions.

show a slight decline from the BAU, 10%, 20% and 30% renewable energy penetration scenarios and this can be attributed to the replacement of thermal generation sources, which usually have high availability, with renewable energy sources with much lower availability in the development of the scenarios. The proportion of electricity produced from renewable energy sources in GWh increased from 0.08% in the BAU scenario to 2.73%, 5.69% and 8.04% for the 10%, 20% and 30% renewable energy penetration scenarios respectively. This further strengthens the fact that, with the right policy measures in place, the country can diversify its electricity generation and move away from the heavy dependence on thermal generation sources with its attendant environmental issues to more renewable energy-based generation.

### *GHG emission analysis*

Ray & De [36] observed that energy management (planned, efficient, and prudent use of electricity) and expanded use of renewable energy have been perceived to be the two most suitable choices for GHG emissions reduction. The study revealed that Ghana is capable of using renewable energy to move towards a low-carbon economy. Ghana's GHG emissions reduction capacity illustrated in Fig. 5, shows a consistent reduction in GHG emissions in BAU, 10%, 20%, and 30% renewable penetration scenarios respectively. Annual GHG emissions are expected to increase to 7928.8 thousand $MtCO_2$ eq, 6771.9 thousand $MtCO_2$ eq, 5374.2 thousand $MtCO_2$ eq and 3919.1 thousand $MtCO_2$ eq for the BAU, 10%, 20%, and 30% renewable penetration scenarios.

GHG emissions from BAU increased cumulatively, from 1665.2 thousand $MtCO_2$ eq in 2016 to 52,545.2 thousand $MtCO_2$ eq in 2030 as shown in Figure SM3 (supplementary material). The renewable energy penetration scenarios, however, saw a decline in GHG emissions from the BAU value of 52,545.2 thousand $MtCO_2$ eq in 2030 to 46,178.3 thousand $MtCO_2$ eq 41,603.2 thousand $MtCO_2$ eq and 38,332.6 thousand $MtCO_2$ eq in 2030 for the 10%, 20%, and 30% renewable penetrations scenarios respectively. These translate into 12.1%, 20.8% and 27.0% reductions of GHG emissions by 2030 from the BAU scenario. The comparison of the GHG emissions from all the scenarios presented in Figure 9 shows a significant reduction in the GHG emissions largely due to the replacement of 10% 20% and 30% of thermal generation sources with equivalent amounts of renewable energy sources. The higher the amount of renewable energy in the mix, the higher the GHG emissions reduction potential as seen in the comparison from Figures SM4a, SM4b, SM4c and SM4d (supplementary material).

The Cumulative GHG emissions shown in Figure SM4 suggests that Ghana can unconditionally reduce its GHG emissions in the electricity sector by 15% compared to the BAU scenario with a little over 10% of renewable energy penetration, in line with Ghana's Intended Nationally Determined Contribution (INDC). In a related report, Kusumadewi et al. [37] concluded that renewable energy sources used in electricity generation are highly useful in long-term planning of power generation and GHG emissions mitigation. Liu et al. [38] also intimated that production of renewable energy would not only decrease the emission of $CO_2$ from the electricity sector, but also serve as a significant catalyst for energy stability. Therefore, it is critical for all stakeholders to accelerate Ghana's renewable energy deployment policies in order to reduce GHG emissions as a primary climate mitigation policy in the power sector, as the study offers a practical solution for combating climate change locally and globally. Ghana's renewable energy goals align with the Africa Union's Agenda 2063 and SDGs, therefore, putting measures to reducing GHG emissions will work towards achieving both continental and global goals of Agenda 2063 and SDGs respectively. Specifically, achieving Agenda 2063 goal 7, which states that "environmentally sustainable and climate resilient economies and communities" and help towards the achievement the following targets; sustainable natural resource management and Biodiversity conservation, sustainable consumption and production patterns, climate resilience and natural disasters preparedness and prevention and finally renewable energy as well as SDG goals and targets 7, 7.1, 7.2, 7a, 11.7, 11.b, 12.2, and 13.2 By demonstrating the feasibility and benefits of renewable energy integration, we pave the way for other African countries to follow suit and collectively work towards a sustainable and resilient future.

## Conclusions

The results show that renewable energy integration into the electricity generation mix could reduce the GHG emissions in the electricity generation sector if it replaces fossil fuels used for electricity generation. In order to keep up with the fast population growth rate and the country's increased economic activity, it is unavoidable that renewable energy will be used to compensate for the increased demand for electricity, especially in the country's rapid development, taking into account the adverse effects of climate change resulting from the continued reliance on fossil fuels. Again, we have demonstrated that by achieving renewable energy penetration in the electricity generation mix, GHG emissions could be significantly reduced in the electricity sector. These findings are crucial for countries aiming to fulfil their climate commitments and reduce their carbon footprint. Again, given that Ghana's commitments align with the Africa Union's Agenda 2063, our research provides valuable policy insights for other African countries striving to achieve sustainable development and climate goals. The strategies and lessons learned from our study can be extrapolated to other African nations.However, for Ghana to achieve a successful generation mix as indicated by this study, the country will have to progressively work towards the elimination of some identified barriers such as Market design problems that obstruct the integration of renewables, cost of financing high-interest rates, lack/Inadequate access to finance and long-term capital, grid connection constraints and lack of grid capacity, etc. as identified by Mahama et al. [17]. Ghana therefore must take advantage of the global market of renewable energy penetration to learn and transfer some appropriate technologies and develop new policies and regulations for rapid renewable energy development.

The penetration of renewable energy is not just beneficial for electricity generation but also the ability to reduce GHG emissions to ensure sustainable development and reduce the impacts of climate change resulting from the electricity sector. Therefore, Government's intervention in the need to introduce appropriate policies to help boost the renewable energy markets is needed, since there is currently little competition between renewables and conventional energies. The implementation of the recently developed Renewable Energy Master Plan of Ghana is very key to the realization and achievement of various renewable energy targets. Again, government and stakeholders should build the local capacity of technical experts to handle the development, operation and maintenance of renewable energy projects from feasibility to decommissioning. Training centres should be established purposely for renewable energy technologies.

## CRediT authorship contribution statement

**Ebenezer Nyarko Kumi:** Conceptualization, Methodology, Data curation, Software, Visualization, Writing – original draft. **Mudasiru Mahama:** Conceptualization, Methodology, Data curation, Visualization, Writing – review & editing.

## Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.sciaf.2023.e01843.

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